ComfyUI Template Agent

Research Phase Demo: Agent-Powered Template Creation

4 production templates · **6** agent skills · **5.5M+** combined downloads

Alvdansen Labs · #proj-comfy-agent · April 2026

The Problem

ComfyUI has **400+ workflow templates** and **8,400+ custom nodes** in its registry.

Creating a new template today requires:

1. **Manual discovery** — browsing the registry for uncovered node packs
2. **Compatibility research** — checking cloud support, API auth, model formats
3. **Workflow authoring** — hand-writing JSON with correct type connections
4. **Guideline compliance** — manually checking 12 submission rules
5. **Documentation** — writing index.json metadata and submission docs from scratch

Each template takes **hours of manual work** across multiple tools and references.

The Solution

An **agent toolkit** that guides template creators through the entire pipeline:

```
Discover → Audit → Validate → Compose → Document
```

Six Claude Code skills, each handling one phase of template creation.

One orchestrator skill (`/comfy-flow`) chains them into a single guided session.

> "What should we build next?" → "Here's a submission-ready template" in one session.

Architecture

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Three layers:

- **Agent Skills** — 6 slash commands invoked in Claude Code
- **Python Modules** — httpx, Pydantic, DiskCache powering each skill
- **Data Sources** — Registry API, GitHub templates repo, ComfyUI Cloud MCP

Skill Pipeline

Step	Skill	What It Does
1	`/comfy-discover`	Surfaces trending, new, and popular nodes from the registry
2	`/comfy-template-audit`	Searches existing templates, finds coverage gaps
3	`/comfy-validate`	Checks workflow JSON against 12 submission guidelines
4	`/comfy-compose`	Builds type-safe workflow JSON from scratch or scaffold
5	`/comfy-document`	Generates index.json, Notion markdown, thumbnail spec

`/comfy-flow` orchestrates all five steps in sequence with context passed between phases.

Discover & Audit

Finding What to Build

`/comfy-discover` queries the registry API for:

- Trending nodes (by recent downloads)
- New nodes (recently published)
- Category filters (video, audio, image, 3D)
- Random picks for inspiration

`/comfy-template-audit` then checks:

- Is this node pack already covered by a template?
- What are the highest-download uncovered packs?
- Which categories have the fewest templates?

Validate & Compose

Building the Workflow

`/comfy-compose` builds workflow JSON with:

- Type-safe node connections (input/output type matching)
- Scaffold mode (start from existing template, adapt)
- From-scratch mode (graph builder with layout engine)

`/comfy-validate` checks against 12 rules:

- Cloud compatibility (no local paths, supported models)
- API node auth requirements (7 providers detected)
- Submission guidelines (naming, structure, metadata)

> Result: a valid `workflow.json` ready for submission.

Document & Submit

Generating Submission Materials

`/comfy-document` produces:

- `index.json` — Template registry metadata (media type, models, custom nodes, IO spec)
- `submission.md` — Notion-formatted markdown for the submissions database
- **Thumbnail spec** — Dimensions, format requirements, suggested content

All extracted automatically from the workflow JSON.

> Result: a complete submission package — workflow, metadata, and documentation.

Templates Built

Template	Node Pack	Downloads	Status
MelBandRoFormer Audio Separation	comfyui-melbandroformer	240K+	Submitted
Florence2 Vision AI	comfyui-florence2	1.25M+	Demo
GGUF Quantized txt2img	ComfyUI-GGUF	1.69M+	Demo
Impact Pack Face Detailer	comfyui-impact-pack	2.37M+	Demo

Coverage:

- 3 media types (image, audio, vision)
- Node complexity from 5 (linear) to 11 (fan-out)
- Combined **5.5M+ downloads** across targeted node packs

Metrics

Metric	Value
Skills delivered	6
Templates created	4
Validation rules	12
Registry nodes indexed	8,400+
Existing templates searchable	400+
API providers with auth detection	7
Test suite	212 tests, ~0.5s

Development velocity:

- v1.0 (6 skills): 14 plans, ~1 hour total execution
- v2.0 (4 templates): built using the skills themselves

What We Learned

1. **Registry API is the key enabler** — node metadata (inputs, outputs, types) makes type-safe composition possible
2. **Scaffold > from-scratch** for most templates — starting from a similar workflow and adapting is faster than building node-by-node
3. **Validation catches real issues** — GGUF model format warnings, API auth requirements, and cloud compatibility flags prevented submission errors
4. **The skill pipeline is the product** — individual skills are useful, but the guided `/comfy-flow` session is where the value compounds

Next Steps

Shipped ✓

- Published at alvdansen/comfyui-template-agent
- GitHub Actions CI (ruff + pytest), demo GIF, full documentation
- 4 templates with submission packages ready for review

Research Phase Integration:

- Submit 4 templates to `workflow_templates` repo
- Deeper MCP Server v0.2.1 integration for live cloud validation
- Expand coverage: video, ControlNet, IPAdapter node packs

Roadmap:

- Template auto-generation from node pack metadata alone